Comparison of Zr, Bi, Ti, and Ga as Metal Contacts in Inorganic Perovskite $CsPbBr_3$ Gamma-Ray Detector

Lei Pan , Yuanxiang Feng , Jinsong Huang , Ayub Sherif, and Lei R. Cao , Senior Member, IEEE

***Abstract*— We report a performance comparison of Zr, Bi, Ti, and Ga as metal contacts on solution-grown $CsPbBr_3$ single-crystal gamma-ray detectors. The same $CsPbBr_3$ single crystal was repeatedly deposited with different metal contacts for the evaluation of gamma spectral performance to eliminate crystal-to-crystal variation. Leakage current as low as 2–3 nA at −200 V with detector thickness 2–3 mm was consistently achieved with Zr, Bi, Ti, and Ga as Schottky contact metal, which is comparable to the leakage current level of a commercial cadmium zinc telluride (CZT) detector. Gamma energy spectra with a prominent photopeak at 662 keV were consistently acquired with all studied metal contacts. The best resolution at 662 keV is 11% with a peak-to-Compton valley ratio of 2, indicating an improved spectrum shape than previous results. Through Americium-241 (^{241}Am) alpha spectra acquisition, the hole mobility of the $CsPbBr_3$ single crystal is evaluated to be 1.78 cm2 V^{-1} s^{-1}.**

We also discussed issues related with the $CsPbBr_3$ gamma detector architecture and interface layers, commonly encountered during $CsPbBr_3$ gamma detector fabrication, that is, impact of electron/hole transport layer and surface oxide layer on gamma detector spectral performance, comparison of different metal–perovskite–metal structures. The persistent $CsPbBr_3$ gamma spectral performance marks the good reproducibility of $CsPbBr_3$ gamma detectors, further enabling $CsPbBr_3$ detector field applications.

***Index Terms*— Gamma energy spectrum, gamma-ray detector, inorganic perovskite $CsPbBr_3$, metal contacts.**

I. Introduction

ROOM-TEMPERATURE semiconductor gamma-ray detectors with good energy resolution have wide applications in nuclear defense, homeland security, medical imaging, and academic research [1]–[3]. Recently, the lead (Pb) halide perovskites emerged as a promising alternative to the cadmium zinc telluride (CZT) detector owing to their favorable properties for room-temperature X- and gamma-ray detection. The band-gap energy of lead halide perovskites (1.5–3.1 eV [4]–[6]) is high enough to reduce thermal noise for room temperature operation, but not overly high so that the Fano noise is still small because a larger number of e–h pairs can be generated for the same incident radiation energy. Their high effective atomic number (Z_{eff}) offers a high gamma-ray attenuation coefficient. For example, Z_{eff} of $CsPbBr_3$ is 65.9 with lead constituting 35.7% in weight. Their mobility–lifetime ($\mu\tau$) product has been shown in the range 10^{-3}–10^{-2} cm^2/V [4], [6], with still potentials for improvement. More important, the perovskite raw material cost is low (estimated raw material cost of \$0.3–3 cm^{-3} after scaling up [4]), and the growth method is not restricted to thin-film format [4], [7]–[9], which offers large volume needed to intercept highly penetrating gamma-rays. Among the perovskites, the all-inorganic $CsPbBr_3$ attracts special attention due to its superior long-term chemical and mechanical stability against the organic–inorganic perovskites [8], [10], [11]. Gamma spectral resolution of 3.8% resolution at 662 keV has been reported from the $CsPbBr_3$ detector made of melting-grown single crystals [12]. We previously reported 5.5% resolution at 662 keV for the $CsPbBr_3$ detector fabricated from a solution-grown single crystal [13]. This work focuses on their consistent spectroscopy performance with an improved gamma spectral feature (i.e., peak-to-Compton valley at 2) when applying different metal contacts.

The $CsPbBr_3$ detector resolution performance is determined by $CsPbBr_3$ single crystal quality as well as the detector architecture and fabrication process. A clearly understood architecture is important to achieve good reproducibility for a perovskite gamma detector. In solar cell applications, the perovskite layer is always sandwiched between an electron transport layer (ETL) and a hole transport layer (HTL) to improve the charge extraction [14]–[16]. Naturally, the ETL/HTL is also inherited into perovskite device architecture aimed for X-ray and gamma-ray detection in early studies [4], [7], [17], [18]. With the underlying theory not widely discussed, our analysis concludes that the addition of ETL/HTL in gamma-ray detection might not be as necessary as it is in solar cell applications. Finally, it is well known that the native oxide layer, formed intentionally or unintentionally, on semiconductor surfaces could have a significant impact on device performance. Hence, it is important to investigate the $CsPbBr_3$ surface for a native oxide layer and understand its potential influence on detector performance.

In this work, we report $CsPbBr_3$ gamma-ray detector spectral performance reproduced with various Schottky contact metals, that is, zirconium (Zr), bismuth (Bi), titanium (Ti), and gallium (Ga). To eliminate the influence of crystal-to-crystal

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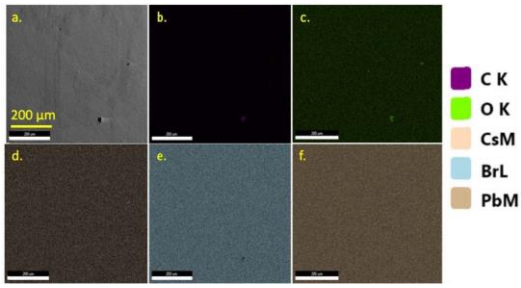

Fig. 1. $CsPbBr_3$ surface topography and chemical composition after polishing and cleaning. (a) $CsPbBr_3$ surface topography taken by SEM. Element overlay of (b) carbon, (c) oxygen, (d) cesium, (e) bromine, and (f) lead.

quality variation, only one solution-grown $CsPbBr_3$ single crystal (labeled as crystal #1) was deposited repeatedly by all studied metal electrodes for gamma spectral performance comparison. For topography evaluation of metal contact and $CsPbBr_3$ detector stability testing, however, other $CsPbBr_3$ crystals were used. We used the metal–perovskite–metal (MPM) structure for our $CsPbBr_3$ detector architecture and presented a theoretical comparison of different MPM structures in a radiation detector. Leakage current as low as 2–3 nA at 200 V reverse bias was consistently achieved with Zr, Bi, Ti, and Ga as metal contact on the same $CsPbBr_3$ single crystal #1, which is comparable to the leakage current of a commercial CZT detector. Gamma spectra with a distinguished 662 keV peak were consistently acquired with different metal electrodes. The best resolution acquired is 11% at 662 keV with a peak-to-Compton valley ratio of 2. Americium-241 (^{241}Am) alpha energy spectra were also acquired to evaluate the hole mobility for the $CsPbBr_3$ single crystal #1. The fact that the same $CsPbBr_3$ single crystal can be used repeatedly with consistent gamma spectral performance indicates the robustness and good quality of reutilization of $CsPbBr_3$ single crystals.

II. Performance of the $CsPbBr_3$ Detectors

A. $CsPbBr_3$ Gamma Detector Fabrication

The as-grown $CsPbBr_3$ single crystal grown from solution was polished by diamond polishing paste and cleaned by toluene, followed by electrode deposition. A scanning electron microscope (SEM) image shows a typical $CsPbBr_3$ surface topography after polishing and cleaning [see Fig. 1(a)]. The dark spots on the surface are the organic residues from polishing paste where carbon (C) and oxygen (O) are concentrated, as evidenced by the surface chemical composition analysis using SEM [see Fig. 1(b) and (c)]. Apart from its concentration on organic residues, oxygen also exists uniformly on $CsPbBr_3$ surface [see Fig. 1(c)], indicating the formation of a native oxide layer on the surface under exposure to ambient air. The analysis of the impact of an oxide layer on the detector spectral performance is discussed later in this work. After surface polishing and (unintentional) passivation by a native oxide layer, the metal contacts were deposited on $CsPbBr_3$ surface. Zr, Ti, and Au were deposited using electron-beam evaporation, and Bi was deposited by thermal evaporation. Ga was simply brushed onto $CsPbBr_3$ surface. An additional Au layer was deposited on top of Zr, Ti, or Bi to protect metals from oxidation and scratching. The material properties of interest for the current discussion are listed in Table I.

TABLE I

Material Properties Related to Detector Fabrication

	Work Function ϕ_m (eV)	Melting point (°C)	Linear thermal expansion coefficient (µm/(m·K)) at 25 °C
Zr	4.05	1855	5.7
Ti	4.33	1668	8.6
Bi	4.31	271	13.4
Ga	4.32	29.7	18
Au	5.10	1064	14.2
$CsPbBr_3$	Ec=-3.3 eV Ev=-5.6 eV [12]	567	~40 [9]

Ec stands for conduction band minimum. Ev stands for valance band maximum.

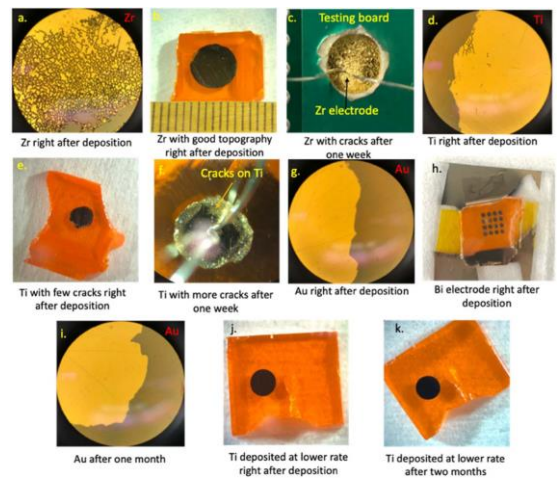

Fig. 2. Contact quality of various metals on $CsPbBr_3$. (a) Zr with cracks right after deposition. (b) Zr with good topography right after deposition. (c) Zr after testing for one week. (d) Ti with less cracks right after deposition. (e) Ti with few cracks right after deposition. (f) Ti after testing for one week. (g) Au with no cracks right after deposition. (h) Bi with no cracks right after deposition. (i) Au with no cracks after one month. (j) Ti with no cracks right after deposition at optimized low deposition rate. (k) Ti with no cracks after two months at optimized low deposition rate.

B. Metal Contact Topography Quality Comparison

A low work function (WF) metal must be used to form a Schottky barrier with p-type $CsPbBr_3$. Our previous work and other studies in literature have used Ga as Schottky contact metal on perovskites [12], [13], [19]–[21]. Although Ga is satisfactory for detector testing in a laboratory environment, it is not suitable for field applications due to its low melting temperature (29.7 °C). Obviously, when temperature is higher than 29.7 °C, Ga will melt and deform, resulting in a poor Schottky contact quality. To improve $CsPbBr_3$ detector performance with respect to energy resolution and stability, other low WF metals, that is, Zr, Bi, and Ti were evaluated and their performance was compared.

Zr was selected due to its favorable lower WF (4.05 eV) than that of Ga, which, in principle, would lead to a higher Schottky barrier height (SBH) with a p-type $CsPbBr_3$ and subsequently a lower bulk leakage current. However, we found that a good Zr electrode deposition on $CsPbBr_3$ could be rarely obtained. Exfoliations and cracks usually appeared on Zr electrode right after deposition [see Fig. 2(a)]. Even when a Zr electrode was

made with good topography immediately following a well-controlled deposition process, cracks will gradually build up over time and eventually cause a significant degradation of the electrode quality [see Fig. 2(b) and (c)]. Ti was tested as it has similar chemical properties with Zr but a lower melting point. Compared to Zr, the Ti electrode was typically achieved with much less cracks right after deposition [see Fig. 2(d)], though cracks would eventually build up on the electrode (e.g., after a week) [see Fig. 2(e) and (f)]. We tested Bi as it is reported chemically inert on perovskites [22]. Bi and Au electrodes were always obtained without any cracks after deposition and stayed intact in ambient air for at least one month [see Fig. 2(g)–(i)].

The differences in topographic behavior of the metal electrodes are believed to be a consequence of mismatch between thermal expansion coefficients and the thin-film thermal stress. The electrode cracking is likely to happen during the cool down process after deposition. The higher the deposition temperature, the more is the subsequent thermal stress. Zr has the highest melting point (1855 °C) and hence the largest amount of cracks. Ti has a lower melting point (1668 °C) than Zr and thus yields less cracks. Au (1064 °C) and Bi (271.4 °C) have even lower melting point than Ti and hence a negligible thermal stress. It can be inferred that the cracks on a certain metal electrode can be minimized by decreasing the deposition temperature through reduced deposition rate. By doing so, a crack-free Ti electrode on $CsPbBr_3$ was successfully achieved [see Fig. 2(j)], which remained crack-free for at least two months in ambient air [see Fig. 2(k)]. Contrary to Zr, Ti, and Bi contacts, Ga has the advantage of forming a contact with $CsPbBr_3$ in liquid phase, thanks to its good wetting ability [23], which circumvents the thin-film thermal stress issue at high temperature. Thus, Ga Schottky contact enables a reliable and stable performance of the $CsPbBr_3$ detector at a constant room temperature, while Bi and Ti are good candidates for $CsPbBr_3$ detectors in field applications due to their stability on $CsPbBr_3$.

C. Gamma Spectral Performance of Zr, Bi, Ti, and Ga

To eliminate the influence of crystal-to-crystal variation, we repeatedly applied different metal contact on one $CsPbBr_3$ crystal (crystal #1) for gamma spectral performance comparison. We washed away the metal contacts of previously fabricated device, polished, cleaned, and redeposited another metal contact on the same crystal (#1) with a decreasing thickness due to multiple polishing. The testing sequence of metal electrodes was Zr, Bi, Ti, and Ga. $CsPbBr_3$ with deposited metal electrode was stored and tested in ambient air. The $CsPbBr_3$ detector with Zr metal contacts ($Zr/CsPbBr_3/Zr$) is designed to have a Zr planar electrode, on the one side, and a Zr pixelated electrode, on the other side, which formed a double Schottky structure. The pixelated electrode structure introduces the small pixel effect that allows for the improvement of detector resolution [24], [25]. The $CsPbBr_3$ detector with Bi electrodes ($Bi/CsPbBr_3/Bi$) followed the same double Schottky and pixelated electrode structure design as with Zr. The $CsPbBr_3$ detector with Ti as Schottky metal ($Ti/CsPbBr_3/Au$) had a planar Ti electrode, on the one side, and a planar

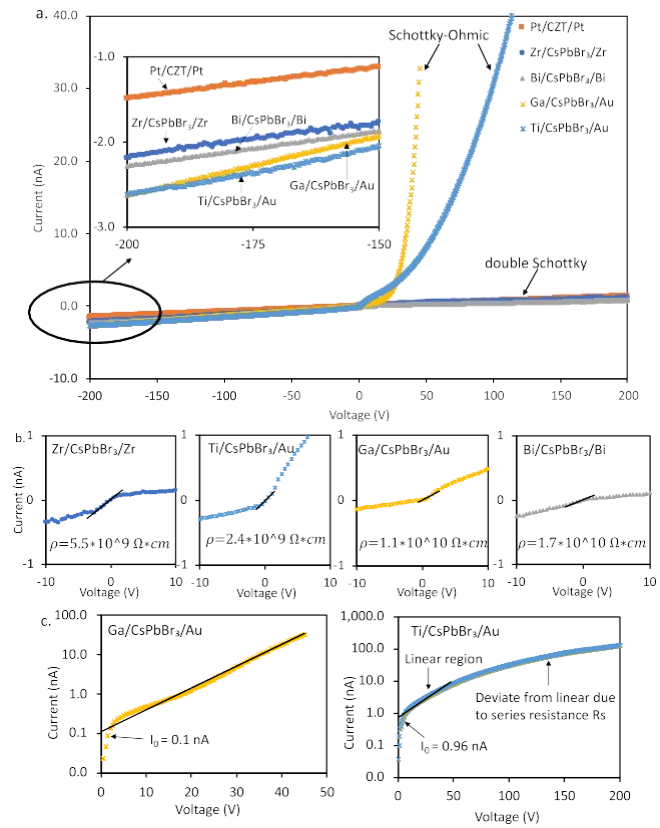

Fig. 3. I –V curves of $CsPbBr_3$ crystal #1 detector with different metal contacts. (a) I –V curves in ±200 V range. (b) I –V curves zoomed in to show the ±10 V range. (I –V curve in ±1 V range was used to calculate the resistivity for each metal contact scheme and the resistivity of $CsPbBr_3$ crystal #1 $\rho \approx 10^9 \times h$ cm is obtained by taking an average of all resistivities calculated.) (c) Log(I)–V plot linear fitting for Schottky diode reverse saturation current I_0 determination. Diode area A of $Ga/CsPbBr_3/Au$ and $Ti/CsPbBr_3/Au$ detectors are $\pi\ (0.5 \times 10^{-3})^2$ and $\pi\ (1.3 \times 10^{-3})^2$ m^2, respectively.

Au electrode, on the other side, which yielded a Schottky–Ohmic structure. The $CsPbBr_3$ detector with Ga Schottky metal design ($Ga/CsPbBr_3/Au$) had a planar Ga electrode, on the one side, and a pixelated Au electrode, on the other side, yielding a Schottky–Ohmic structure. For the $CsPbBr_3$ detectors with pixelated electrode design, the current–voltage (I –V) curve and gamma spectrum were acquired with only one pixel biased and all other pixels floating. The detector structures and dimensions with different metal electrodes are summarized in Table II.

The designed $CsPbBr_3$ detector configurations with respect to current-rectifying behavior were validated both experimentally and theoretically. The I –V characteristics measured of $CsPbBr_3$ crystal #1 detector with different metal electrodes agree well with corresponding designed configurations [see Fig. 3(a)]. As expected, the Schottky–Ohmic structure of $CsPbBr_3$ detectors with Ti and Ga as Schottky electrodes exhibits a current blocking effect under reverse bias (applied electric field direction from Ti or Ga to $CsPbBr_3$) and no current blocking under forward bias. In comparison, the double Schottky structure of $CsPbBr_3$ detectors with Zr and Bi as electrodes shows a current blocking effect under both biasing directions. The I –V curves in Fig. 3(a) were further zoomed in to show the current behavior in

TABLE II

STRUCTURES AND DIMENSIONS OF $CsPbBr_3$ CRYSTAL #1 DETECTORS WITH DIFFERENT METAL ELECTRODES

	Structure and layer thickness	Electrode area
Zr	Zr(150nm)/$CsPbBr_3$(~3mm)/ Zr(150nm)	Planar Zr ø=5.5mm Pixelated Zr ø=1mm
Ti	Ti(120nm)/$CsPbBr_3$(~2.3mm) /Au(50nm)	Planar Ti ø=2.6mm Planar Au ø=2.6mm
Bi	Bi(250nm)/$CsPbBr_3$(~2.7mm) /Bi(250nm)	Planar Bi ø=5.5mm Pixelated Bi ø=1mm
Ga	Ga(2mm)/$CsPbBr_3$(~2mm)/A u(120nm)	Planar Ga ø=3.5mm Pixelated Au ø=1mm

An extra 30-50 nm Au layer is deposited on top of Zr, Ti and Bi to protect them from oxidation and scratching. "ø" stands for diameter. Crystal #1 thickness decreased as consequence of polishing for detector re-fabrication of each metal electrode.

the ± 10 V range [see Fig. 3(b)]. It can be seen that the $I-V$ curve slope is reduced, as bias increases in both reverse and forward directions for Zr/$CsPbBr_3$/Zr and Bi/$CsPbBr_3$/Bi detectors. Meanwhile, the $I-V$ curve slope of Ti/$CsPbBr_3$/Au and Ga/$CsPbBr_3$/Au detectors decreases in the reverse bias direction and increases in the forward bias direction. The resistivity of the $CsPbBr_3$ crystal #1 was calculated to be about $\rho = 9 \times 10^9\ \Omega\ \mathrm{cm}$.

Based on energy band bending theory, $CsPbBr_3$ fermi level relative to metal electrode WF ρ_m was calculated, similar to the analysis performed on CdTe crystals in [26], with results also agreeing with the current-rectifying behavior of the designed $CsPbBr_3$ detector configurations. $CsPbBr_3$ shows p-type behavior [12] and its fermi-level position E_{Fs} above valance band maximum E_v can be calculated by $(kT/q)\ln(N_v/p)$, where k is Boltzmann's constant, T is the room temperature 293 K, q is the electronic charge, $N_v = 2(m_p kT/2\pi\hbar^2)^{3/2}$ is the effective density of states in valance band, and $p = 1/q\rho\mu_p$ is the hole concentration. For $CsPbBr_3$, we know the hole and electron density of states effective mass $m_p = m_n = 0.23m_0$ [9], where m_0 is the electron mass. Hole mobility μ_p of $CsPbBr_3$ crystal #1 was evaluated later in this work to be 1.78 $cm^2\ V^{-1}\ s^{-1}$. Using these numbers, the hole concentration for $CsPbBr_3$ crystal #1 is calculated to be $3.9 \times 10^8/cm^3$ and the distance between E_{Fs} and E_v is 0.57 eV. Given the conduction band minimum E_c and E_v of $CsPbBr_3$ is 3.3 and 5.6 eV [12], E_{Fs} is 5.03 eV. Metal electrode with ρ_m smaller than 5.03 eV would form a junction with $CsPbBr_3$ whose energy bands bend toward valance band, which means a Schottky contact is formed. Meanwhile, Ohmic contact would be formed when the metal electrode ρ_m is larger than 5.03 eV. As shown in Table I, Zr, Bi, Ti, and Ga have ρ_m smaller than 5.03 eV as opposed to Au with ρ_m larger than 5.03 eV, which theoretically supports that a Schottky contact is formed between Zr, Bi, Ti, or Ga and $CsPbBr_3$, while Ohmic contact is formed between Au and $CsPbBr_3$. Besides, the Schottky contact depletion region width W can be calculated as $W = (2\varepsilon_0\varepsilon_r (V_{bi} + V_a)/(qp))^{1/2}$, where ε_0 is the vacuum permittivity, $\varepsilon_r = 29.4$ is the static dielectric constant of $CsPbBr_3$ single crystal [27], V_{bi} is the built-in potential of Schottky barrier, and V_a is the applied reverse bias. For a $CsPbBr_3$ detector with a thickness of 3 mm, it will be fully depleted at a reverse bias of several hundred volts.

SBH of Ga/$CsPbBr_3$ and Ti/$CsPbBr_3$ contacts were extracted from the $I-V$ curve measured with Ga/$CsPbBr_3$/Au and Ti/$CsPbBr_3$/Au detectors shown in Fig. 3(a), respectively. The current in a Schottky diode according to thermionic emission can be found as [28]

$$I = I_0 \exp\frac{qV}{nkT}\left(1 - \exp\left(-\frac{qV}{nkT}\right)\right) \quad (1)$$

$$I_0 = AA^*T^2 \exp(-q\rho_{B0}/kT) \quad (2)$$

where I_0 is the reverse saturation current, V is the applied bias voltage, n is the ideality factor, A is the diode area, and $A^* = 4\pi q m_p k^2/h^3$ is the effective Richardson constant where h is the Planck constant. ρ_{B0} is the SBH at zero bias. I_0 is obtained from the intercept to the y-axis of the linear fitting to the linear region of the forward bias $\log(I)-V$ plot [see Fig. 3(c)] and then ρ_{B0} is determined from (2). The SBH of Ga/$CsPbBr_3$ and Ti/$CsPbBr_3$ contacts were calculated to be 0.83 and 0.82 V. It can be seen from Fig. 3(c) that the forward $I-V$ of Ti/$CsPbBr_3$/Au detector deviates from linear behavior at high voltage region, which is known as the effect of a series resistance R_s of the Schottky diode. SBH can be extracted by Cheung and Cheung [29] model with a series resistance R_s taken into account, yielding SBH of Ga/$CsPbBr_3$ and Ti/$CsPbBr_3$ contact to be 0.84 and 0.82 V, respectively, which agrees well with the SBH extracted from (1) and (2). The theoretical SBH of Ga/$CsPbBr_3$ and Ti/$CsPbBr_3$ contacts calculated from $\rho_{B0} = E_g/q - (\rho_m - \chi_s)$ are 1.28 and 1.27 V, respectively (E_g is the band-gap energy and χ_s is the electron affinity). The difference between experimentally measured value and theoretical value of SBH may be due to the effect of surface states or Fermi level pinning.

The leakage current of the $CsPbBr_3$ detector with respective metal electrodes was consistently low in the 2–3 nA range at reverse 200 V [see Fig. 3(a), inset]. This is comparable to the leakage current of a commercial CZT detector of double Schottky structure (Pt/CZT/Pt) with a similar dimension ($4 \times 5 \times 5\ mm^3$) as that of $CsPbBr_3$ detectors [see Fig. 3(a)]. The ^{137}Cs gamma source (5.0 μCi) energy spectra with a prominent peak and/or Compton feature were consistently acquired with $CsPbBr_3$ detectors with all metal electrodes (see Fig. 4). The Ti/$CsPbBr_3$/Au detector with a planar electrode produced a resolution of 11% at 662 keV with an improved peak-to-Compton valley ratio of 2 compared with a smaller peak-to-Compton valley ratio (<2) from our previous results [13]. In comparison, the Zr/$CsPbBr_3$/Zr, Bi/$CsPbBr_3$/Bi, and Ga/$CsPbBr_3$/Au detectors with pixelated electrode structure showed a worse photopeak feature with a smaller peak-to-Compton valley ratio since only one pixel is biased to collect signal, which results in a smaller effective detection volume than the planar Ti/$CsPbBr_3$/Au detector. Energy spectra of 59.5 keV ^{241}Am gamma rays (1.0 μCi) were also acquired by the Ti/$CsPbBr_3$/Au detector [see Fig. 4(e)].

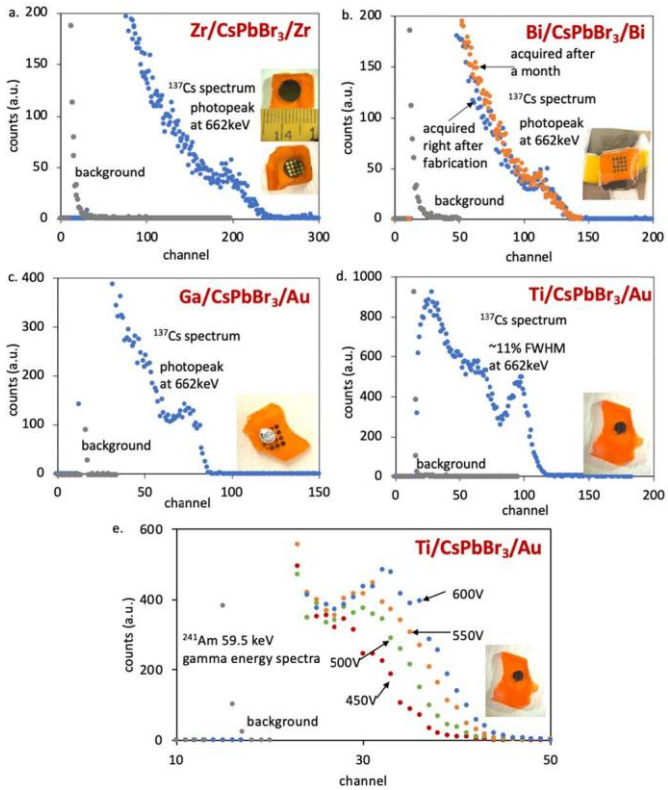

Fig. 4. Gamma energy spectra acquired with $CsPbBr_3$ detectors made of crystal #1 with different metal electrodes. The gamma spectroscopy system used for energy spectra acquisition includes charge-sensitive preamplifier (CAEN MOD A1422F3), shaping amplifier (ORTEC 671), and MCA (ORTEC 926). (a) ^{137}Cs gamma energy spectrum of Zr/$CsPbBr_3$/Zr detector (bias voltage –450 V, shaping 10 μs, acquisition time 2236 s). (b) ^{137}Cs gamma energy spectra of Bi/$CsPbBr_3$/Bi detector (bias voltage –450 V, shaping 10 μs, acquisition time 787 s). (c) ^{137}Cs gamma energy spectrum of Ga/$CsPbBr_3$/Au detector (bias voltage –450 V, shaping 10 μs, acquisition time 2400 s). (d) ^{137}Cs gamma energy spectrum of Ti/$CsPbBr_3$/Au detector (bias voltage –350 V, shaping 10 μs, acquisition time 2318 s). (e) ^{241}Am gamma energy spectra of Ti/$CsPbBr_3$/Au detector (shaping 10 μs, acquisition time 600 s at each bias voltage).

The gamma spectral resolution of Zr/$CsPbBr_3$/Zr and Ti/$CsPbBr_3$/Au detectors degraded one week post measurement due to electrode degradation [see Fig. 2(c) and (f)], while the energy resolution of the Bi/$CsPbBr_3$/Bi detector had not shown any considerable degradation acquired after storage in ambient air for at least one month [see Fig. 4(b)], indicating the stability of Bi contact on $CsPbBr_3$. The Ti electrode, however, has the potential to produce stable spectral resolution as Ti also shows stability on $CsPbBr_3$ surface after optimization as shown earlier [see Fig. 2(j) and (k)]. The $CsPbBr_3$ detector stability under continuously applied reverse bias was tested using other $CsPbBr_3$ single crystals. The dark currents measured with a Bi/$CsPbBr_3$/Au detector made of crystal #2 and with a Ga/$CsPbBr_3$/Au detector made of crystal #3 remained stable for hours under constant –200 V bias [see Fig. 5(a)], which shows little or no ion migration issues. Besides, the ^{137}Cs gamma spectra of the Ga/$CsPbBr_3$/Au detector made of crystal #3 acquired at the beginning and end of a 24-h period under continuously applied –200 V bias did not show a significant difference [see Fig. 5(b)], which also indicates the good stability of Ga/$CsPbBr_3$/Au detector under long bias condition.

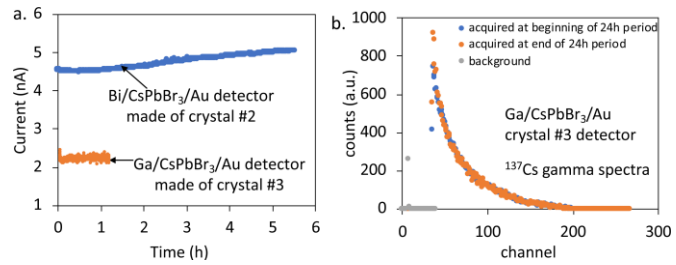

Fig. 5. $CsPbBr_3$ detector stability test. (a) Dark current as a function of time measured with a Bi/$CsPbBr_3$/Au crystal #2 detector (thickness 2.58 mm, planar electrode area: 3.5×3.5 mm^2) and a Ga/$CsPbBr_3$/Au crystal #3 detector (thickness 2.7 mm, planar electrode area: $n\times 2.1\times2.1$ mm^2) under –200 V bias. (b) ^{137}Cs gamma energy spectra acquired with Ga/$CsPbBr_3$/Au crystal #3 detector at the beginning and the end of a 24-h period under continuously applied –200 V (shaping 10 μs, acquisition time 7000 s for each spectrum).

D. $CsPbBr_3$ Alpha Spectra and Hole Mobility Evaluation

Alpha energy spectra were acquired with the Ti/$CsPbBr_3$/Au detector made of the $CsPbBr_3$ crystal #1. The planar Ti electrode was irradiated by a 1.0 μCi ^{241}Am source under reverse bias [Fig. 6(a), inset and (b)], to collect hole-induced signal because essentially all e–h pairs are generated at the surface given the limited penetration depth of α-particles in $CsPbBr_3$ (28.8 μm for 5486-keV α-particle by the stopping and range of ions in matter (SRIM) simulation [30]). Hole-induced ^{241}Am α-energy spectra were collected under different bias voltages [see Fig. 6(a)]. In Fig. 6(a), full energy peaks of the α-particle spectra are significantly broadened because of the 2-cm air gap between the source and the detector. Despite the broadening, the full energy peak clearly shifts to the right with increasing reverse bias and approaches saturation. The ^{241}Am source position was switched to irradiate the Au electrode for electron-induced signal collection [see Fig. 6(d)]. In contrast to the hole-induced spectra, the electron-induced ^{241}Am α-energy spectra indicate inferior peak resolution and smaller pulse height under the same revere bias, which qualitatively suggests a poorer electron $\mu\tau$ than that of holes in $CsPbBr_3$. The hole-induced pulses of ^{241}Am α-particles at the output of charge-sensitive preamplifier were also recorded to evaluate hole mobility, μ, of the $CsPbBr_3$ crystal #1 using the time-of-flight method [see Fig. 6(c) and (e), for an example of the preamplifier waveform]. Rise times of 1000 pulses were used to calculate the average pulse rise time at each bias voltage.

The average hole drift time was estimated by the average pulse rise time under each bias voltage. It can be seen that the carriers drift faster, that is, both average pulse rise time and its deviation decrease, as the applied bias voltage increases [see Fig. 6(e)]. The hole mobility of the $CsPbBr_3$ crystal #1 is evaluated to be 1.78 cm^2 V^{-1} s^{-1}, which is smaller than the mobility of the $CsPbBr_3$ crystal in our previous work (electron–hole averaged mobility $\mu = 22$ cm^2 V^{-1} s^{-1}). This agrees with the inferior gamma spectral resolution of the $CsPbBr_3$ crystal #1 (11% at 662 keV) compared to our previous results (5.5% at 662 keV [13]).

Typically, single charge carrier Hecht equation fitting can be used to evaluate the hole $\mu\tau$ product with full energy peak positions as a function of reverse bias voltage. However, we found that the peak positions in Fig. 6(a) actually significantly

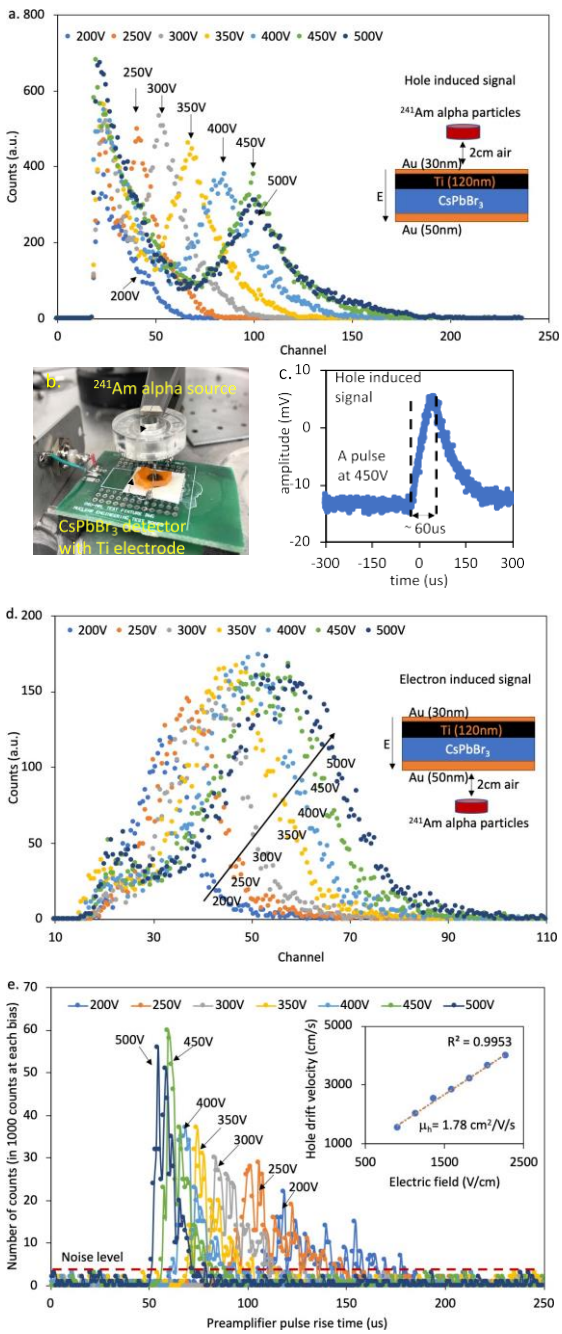

Fig. 6. Ti/$CsPbBr_3$/Au detector made of crystal #1 ^{241}Am alpha energy spectra and hole mobility μ evaluation (spectroscopy system is the same as in Fig. 4). (a) Hole-induced ^{241}Am α-particle energy spectra (shaping 10 μs, acquisition time 600 s at each bias). (b) Experimental setup for hole-induced signal collection. (c) Preamplifier pulse by α-particle hole-induced signal at 450 V. (d) Electron-induced ^{241}Am α-particle energy spectra (shaping 10 μs, acquisition time 600 s at each bias). (e) Hole-induced ^{241}Am α-particle preamplifier pulse rise time histogram. Inset is a fitting for hole mobility evaluation. The hole drift velocity at each bias voltage is calculated using detector thickness divided by average drift time and electric field is calculated using applied bias voltage divided by detector thickness.

deviated from their true positions because of the existence of ballistic deficit in the pulse processing of hole-induced signals, which prevents from an accurate fitting. Ballistic deficit is significant when the pulse rise time is not significantly smaller than the time constants of the pulse processing components used [13]. Fig. 6(e) shows that the smallest pulse rise time is still larger than 50 μs at 500 V. Given the 50 μs decay constant for CAEN MOD A1422F3 preamplifier and 10 μs maximum shaping constant for ORTEC 671 shaping amplifier used in this work, ballistic deficit has affected the peak positions significantly.

III. $CsPbBr_3$ Detector Architecture

A. ETL/HTL in Gamma Detector

The current work and other previously reported studies have experimentally demonstrated a good gamma energy spectral resolution of perovskite detectors without ETL/HTL [12], [13], [19], [20]. The ETL/HTL are semiconductor materials, either inorganic or organic, that meet the criteria for working properly in a perovskite solar cell (PSC), for example, good charge carrier mobility, compatible energy levels with respect to a perovskite absorber [14]–[16]. The ETL/HTL in a PSC generates selectivity for electrons or holes, necessary to drive the charge collection in PSC without an external bias [31]. In a radiation detector where an external voltage is commonly applied for charge extraction, the ETL/HTL is in principle nonessential. One caveat is that a depletion layer can be formed in ETL and perovskite when a junction between ETL (e.g., n-type) and perovskite (e.g., p-type) is reversely biased, which may cause nonlinearity between energy disposition and charge generation. The different band gaps of ETL and perovskites, and thus, disparate e–h creation energies of n-type ETL and p-type perovskites, would lead to different amount of charge being generated from the same energy deposited in the depletion region of ETL and perovskites, which might result in detector energy spectrum resolution degradation. However, the resolution degradation can be negligible under a few practical scenarios such as

1) When the ETL/HTL layer thickness is negligible compared with perovskite crystal thickness.
2) When the doping concentration of the ETL/HTL is high enough so that the depletion region in ETL/HTL is negligible.
3) When Z_{eff} of ETL/HTL is small enough that its interaction with gamma rays is negligible.

When ETL/HTL is used in a perovskite gamma detector, one needs to be aware of the effects of Z_{eff}, doping, and thickness of the ETL/HTL with respect to the potential linearity violation between energy deposition and charge production.

B. MPM Gamma Detector

MPM structure has been used for perovskite gamma detectors with good spectral performance [12], [13], [19], [20]. Here, we present a discussion of the different MPM structures in terms of leakage current performance. In a radiation detector, leakage current has to be as low as possible to warrant a good signal-to-noise ratio for signal collection. For example, the p-type $CsPbBr_3$ detector with high WF metal contacts (e.g., Au) on both sides is a resistive mode detector as there are Ohmic contacts on both sides, which usually has a higher leakage current than the Schottky mode detector with current blocking effect. To form Schottky contact with p-$CsPbBr_3$, a low WF metal must be used on one electrode, while on the other electrode a high WF metal can be used to form a Schottky–Ohmic structure or a low WF metal can be used

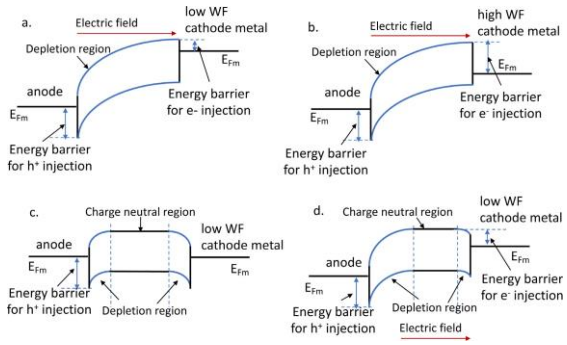

Fig. 7. Energy band diagram of (a) low WF anode metal/p-$CsPbBr_3$ /low WF cathode metal at full-depletion reverse bias. (b) Low WF anode metal/ p-$CsPbBr_3$/high WF cathode metal at full-depletion reverse bias. (c) Low WF anode metal/p-$CsPbBr_3$ /low WF cathode metal at zero bias. (d) Low WF anode metal/p-$CsPbBr_3$/low WF cathode metal at moderate reverse bias.

to form a double Schottky structure. It is important to note that both resistive mode and double Schottky detector produce a symmetric I –V curve, but double Schottky has a current blocking effect while resistive mode not.

The difference of Schottky–Ohmic versus double Schottky structure is shown in Fig. 7. Assuming a full-depletion bias voltage is applied, which is usually the case of a gamma spectroscopy detector, the low WF anode metal blocks hole injection from itself into $CsPbBr_3$ [see Fig. 7(a)]. For the double Schottky, the low WF cathode metal has lower energy barrier for electron injection from itself into $CsPbBr_3$ compared with the relatively higher energy barrier provided by a high WF cathode metal of a Schottky–Ohmic structure [see Fig. 7(a) and (b)]. Hence, the Schottky–Ohmic structure works better theoretically than the double Schottky structure in suppressing bulk leakage current after full depletion is reached. Yet, at moderate reverse bias, the advantage of Schottky–Ohmic structure diminishes because the energy barrier for electron injection from cathode established at thermal equilibrium in double Schottky structure has not been counteracted completely by external reverse bias [see Fig. 7(c) and (d)]. The total leakage current is composed of bulk leakage current and surface leakage current. The latter is surface-specific that depends on surface treatment such as polishing, cleaning, and passivation. Since surface leakage current could be dominant at high reverse bias, in practice the difference between Schottky–Ohmic and double Schottky structure in suppressing total leakage current at high reverse bias may not be significant.

C. Oxide Layer in $CsPbBr_3$ Gamma Detector

As we have seen, a native oxide layer was formed passively on $CsPbBr_3$ surface in ambient air (see Fig. 1). It is known that surface passivation could also be used to form a passivation layer on perovskites surface proactively to improve device performance [17], [32], [33]. The oxide or passivation layer essentially represents a resistive layer between $CsPbBr_3$ and the metal contact. The exact resistance of such layer is material-specific relying on several factors, such as chemical composition, layer thickness, and morphology.

A resistive layer creates a physical isolation between metal and $CsPbBr_3$. Yet, the metal electrode does not have to be

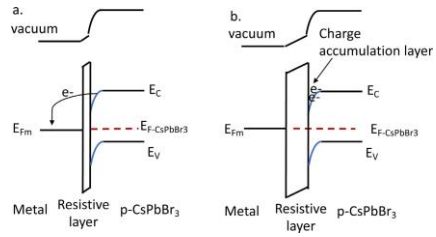

Fig. 8. Energy band diagram of metal/resistive layer/p-$CsPbBr_3$ with (a) resistance small enough allowing for electron tunneling and (b) resistance large enough creating charge accumulation layer at resistive layer/p-$CsPbBr_3$ interface.

in an immediate physical contact with the semiconductor surface to allow charge induction and the linearity between generated charge and induced signal amplitude still hold based on the Shockley–Ramo theorem [34], which is also known as proximity charge sensing that is experimentally demonstrated by Luke *et al.* [35]. If the resistive layer has a resistance small enough so that charge carriers from metal and $CsPbBr_3$ can easily tunnel through its energy barrier [see Fig. 8(a)], no significant charge accumulation occurs at resistive layer–$CsPbBr_3$ interface and its negative effects on gamma spectrum energy resolution could be negligible. On the contrary, if the resistive layer has a resistance large enough, a considerable charge accumulation could occur at the resistive layer–$CsPbBr_3$ interface [see Fig. 8(b)], which could cause polarization and distort the electric field inside the detector and lead to a severe energy resolution degradation. We conclude that an oxide/passivation layer passivates surface defects [36], [37], resulting in an improved energy resolution, but may also introduce charge accumulation that lead to energy resolution degradation. The ultimate effect of oxide/passivation layer on gamma spectrum energy resolution is a balancing act between surface defects passivation and charge accumulation, which depends quantitatively on the exact resistance of oxide/passivation layer and its defect passivation effects. In practice, it is better to maintain tens of nanometers or less for the thickness of oxide/passivation layer during detector fabrication process in order to have a well-controlled detector performance.

IV. Conclusion

We compared the performance of Zr, Bi, Ti, and Ga as metal contacts in a gamma-ray detector made of inorganic perovskite $CsPbBr_3$. The contact topography quality was compared and it was found that thermal stress plays an important role in influencing electrode topography. The Bi electrode stayed stable on $CsPbBr_3$ in ambient air and the resulting gamma spectrum performance has a negligible degradation after storage in air for a month. Ti could also stay stable on $CsPbBr_3$ after deposition process optimization. While Ga is good for making a $CsPbBr_3$ detector, it is mainly confined in a laboratory testing environment. Our work shows that Bi and potentially Ti are suitable for the $CsPbBr_3$ detector field applications. A $CsPbBr_3$ single crystal was used repeatedly for gamma spectral performance evaluation, each time with applying a different metal electrode, to eliminate crystal-to-crystal variation, which indicates the robustness and good recyclability of the $CsPbBr_3$ single crystal. Leakage currents as low

as 2–3 nA at 200 V were consistently acquired by the $CsPbBr_3$ detectors with Zr, Bi, Ti, and Ga as metal contact, which is comparable to that of a commercial CZT detector. Gamma spectra with prominent photopeak at 662 keV were consistently acquired with $CsPbBr_3$ detectors with all studied metal electrodes. The best shaped ^{137}Cs gamma spectrum has a resolution of 11% at 662 keV and a peak-to-Compton valley ratio of 2. ^{241}Am alpha energy spectra were acquired and hole mobility μ of $CsPbBr_3$ crystal #1 was evaluated using time-of-flight method. We also discussed the underlying theories involved with using ETL/HTL in a gamma-ray detector, different MPM structures, and the influence of oxide/passivation layer on a gamma detector spectral performance.

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